COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-DBS-016: Database Developer Service

Document ID	TS-DBS-016
Title	Database Developer Service
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT supported agencies; employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.0
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1. Purpose

OIT is statutorily mandated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy the stakeholder needs surrounding OIT supported databases, the following is submitted as the standard for specifications, methodologies, and practices to provide the service.

The purpose of this document is to communicate the standards that the Governor's Office of Information Technology (OIT) has developed with the intent of providing enterprise standard configurations for Database Developer Service. Standards are based on a best-of-value decision assessment involving multifarious factors, such as stakeholder needs, hardware configurations that are serviced and supported by OIT, as well as operational and security best practices guidelines.

The ultimate purpose of the Database Developer Service is to describe the minimum requirements for designing, installing, securing, monitoring, maintaining, protecting, and decommissioning any OIT supported environment.

2. Scope

These standards apply to OIT and OIT supported agencies. This Standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Additionally, the standards assigned and prescribed by this document, instituted for use by the approval of the OIT Architecture Review Board (ARB), should be used to govern purchase and deployment decisions deemed as “go-forward” or “net-new” strategies or implementations, and are

not intended for use as retroactively enforceable mandate or policy for immediate conformance (especially in cases where cost implications may be prohibitive).

Any/all standards are predicated with the assumption that there can be no point of enforcement unless and until there is an effort to re-procure, renew (contracts), reconfigure (including upgrades), or redevelop the system. At any time that a decision is made to make an investment of resources in sustaining or improving the system, conformance to existing standards should be part of the assessed level of effort for that investment activity. For guidance (or enforcement) in conformance to standards during any of these types of activities, an email request may be sent to OIT_OEA@state.co.us with the subject line "Request for Architecture Review: <projectName/systemName>"

For the purpose of enforcement, this standard is rated CAT 2 according to the Enforcement Category Chart.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Update effective date and/or last review date.
- Publish the ARB approved Standard to the OIT public facing website.

Architecture Review Board (ARB)

- Engage stakeholders and subject matter experts to determine standard.
- Review/Approve Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engage stakeholders and subject matter experts to determine standard.
- Update document standards and version number.
- Present to the Architecture Review Board for approval and adoption.

Additional Assignments

All OIT staff are responsible to be aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Goals

- The Database Development Service ensures that customer databases are built consistently, securely, and accessible to the required application and approved users. This includes implementing requested changes by creating and modifying database Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL) objects, Data Query Language (DQL), and Transaction Control Language (TCL), adding new data points and establishing new security protocols to secure data as well as troubleshooting.

Process

- Database Object Creation and Maintenance: Database Services will be responsible for the creation, update and maintenance of DDL including but not limited to tables, indexes, views, stored procedures, functions, packages, and triggers.
- Database Development and Maintenance: Any database development and database code maintenance will be initiated by the business and worked by database services.
- Database DevOps: Any and all database code will be created, reviewed and maintained in a version control system. (Refer to Database DevOps process)
- Developer Database Privileges: Any developer outside of database services will be granted DML privileges of “select”, “update”, “insert”, and “delete” to development databases unless an exception is given needing a higher level of database access. All DDL changes will be completed by the Database Services team.
- Database Architecture and Design: Database Services will design the database structure and system requirements including the data infrastructure to effectively store the data to be accessed from both internal and external sources. Database Services will analyze the business requirements and produce the data structure to meet these needs including data analysis and effective data flow throughout the system.
- Supported Database Versions: Any database development supported by Database Services must be in the approved supported database versions listed in TS-DBS-001 Supported Databases and Versions.
- Database User Administration: Database user creation and maintenance is defined in the database standard TS-DBS-008 Database User Administration.
- Database Users and Security Maintenance: Controlling, monitoring, auditing user access, and enforcing password requirements to the database will be done by using least privilege as per the requirements listed in TS-DBS-006: Database Security.
- Additional Database Developer Support and Services:
    - Database optimization
    - Query optimization
    - Working with large databases and datasets
    - Index maintenance
    - Performance Tuning
    - ORM Model development and troubleshooting

Requirements

- Database Services will be involved in any new or existing development activity requiring database access and support.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of and approved by the “Owner of Standard” listed in Section 3 (Roles and Responsibilities) of this document.

Requests may be acquired by using the “How-to Guide for Secure Configuration Exception Requests or Policy Exception Requests in ServiceHub” and including the completed policy exception number along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

OIT documents and policies

- Colorado Information Security Policies (CISPs)
- Enforcement Category Chart
- TS-DBS-001 Supported Databases and Versions
- TS-DBS-006: Database Security
- TS-DBS-007: Database Software Management

State documents and policies

- Colorado Brand Guidelines

7. Definitions

Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies. Standards serve as specifications for the implementation of policies and are designed to promote implementation of high-level organization policy rather than to create new policy in themselves.

Production Database: State Agency database used for production purpose to conduct State business.

Non-Production Database: State Agency database used for non-production purpose such as development, testing, staging, demonstration or training.

Business: Employees of supported State agencies, and their respective stakeholders that use DBS supported databases.

DDL: Data Definition Language

DML: Data Manipulation Language

DCL: Data Control Language

DQL: Data Query Language

TCL: Transaction Control Language

Database Object: A data structure used in a database to either store or reference data.

ORM: Object Relational Model

OEA (Office of Enterprise Architecture): The Office of Enterprise Architecture was formed as an office under OIT by Governor's Rule 8 CCR 1501-7.

OIT (Office of Information Technology): The Governor's Office of Information Technology was created and governed by C.R.S. 24-37.5, et seq.

Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception. Typically an exception request is subject to review and approval by an authority in the area.